

Lecture 14 – Ocean Energy Waves & Tides

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Outline

- History of Wave Energy
- Physical Principles and Wave Velocity
- Wave Energy Converters and Power Take-off (PTO) Systems
- Environmental Impact and Economics
- Tidal Energy
- B3: Chap 7 & 9
- B1: Sec 8.3-8.4
- REN21: Renewables 2024 Global Status Report

Current Status of Ocean Energy

- Global technical wave resource: ~2 TW
- Ocean energy contributes <1% of global electricity
- Key challenge: high cost + harsh marine environment
- Huge potential for coastal regions



KEY FACTS FOR 2024

- Ocean power continued to attract investment and support in 2024, reflecting its untapped potential as a predictable renewable energy source.
- The sector has seen a clear shift from pilots and demonstration deployments to full-scale devices and a pipeline of commercial projects.
- With increasing government support, targeted funding, and reforms to regulations and permitting processes, the sector is moving toward scale-up.

513MW total installed capacity

1.6MW of ocean power capacity added

5 developers deployed devices in Europe

Tide vis-à-vis Wave

- **Tides** are caused by the gravitational pull of the Moon and Sun on Earth's oceans, with the Moon having the stronger influence.
- **Waves** are generated by wind blowing across the ocean surface. Wave energy converts the motion of waves into electricity.

Wave

Wind-driven

Seconds-period

Less predictable

Surface
phenomenon

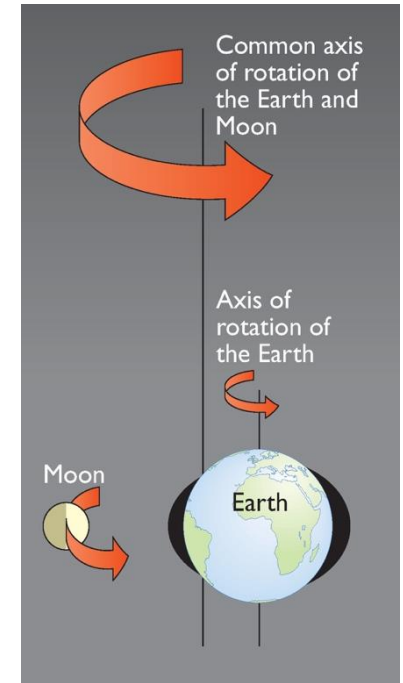
Tide

Gravity-driven

12.4 hr period

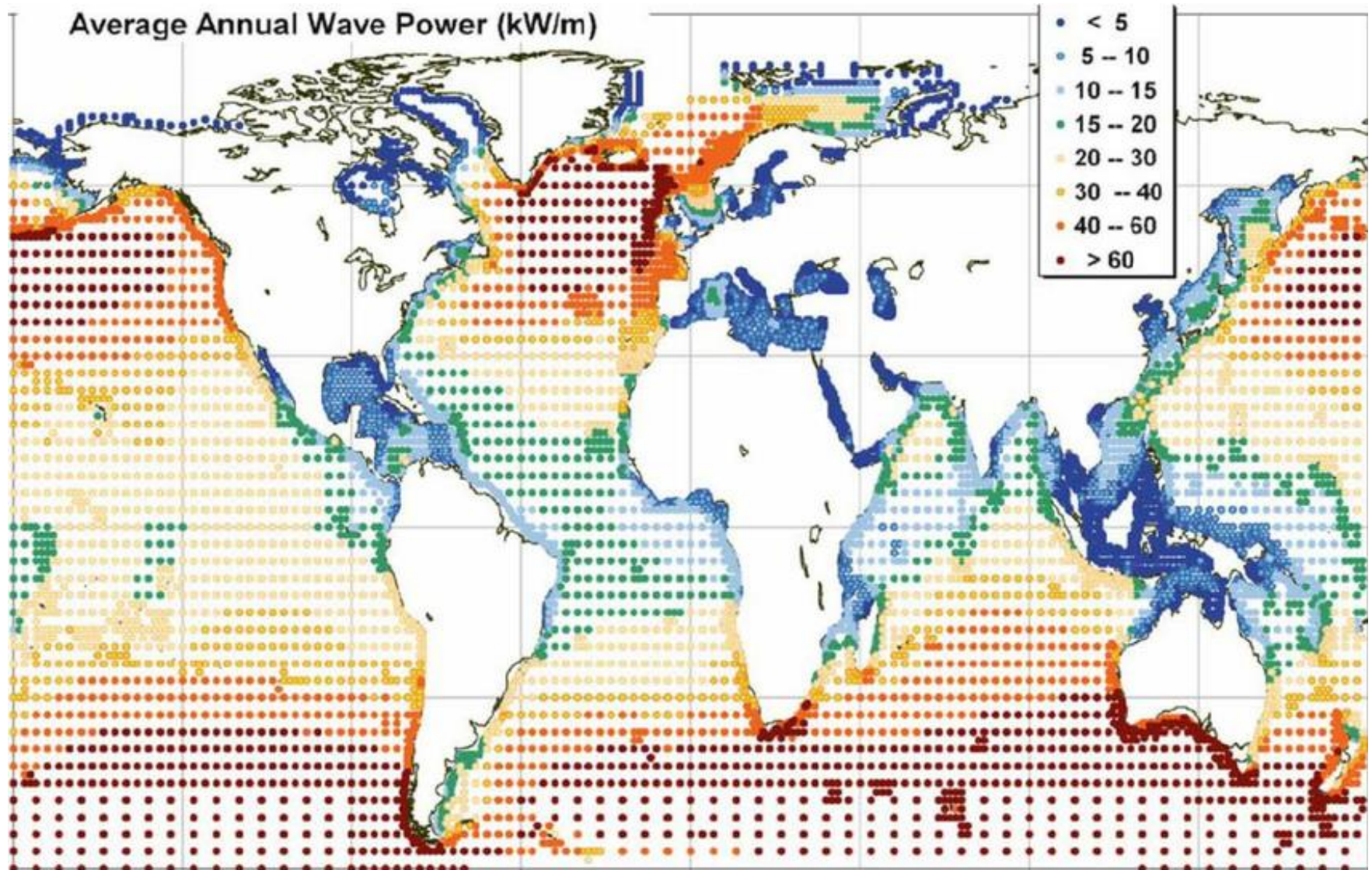
Highly predictable

Whole water body



Wave Energy

Global Annual Average Wave Power



Seasonal Variation in Wave Energy

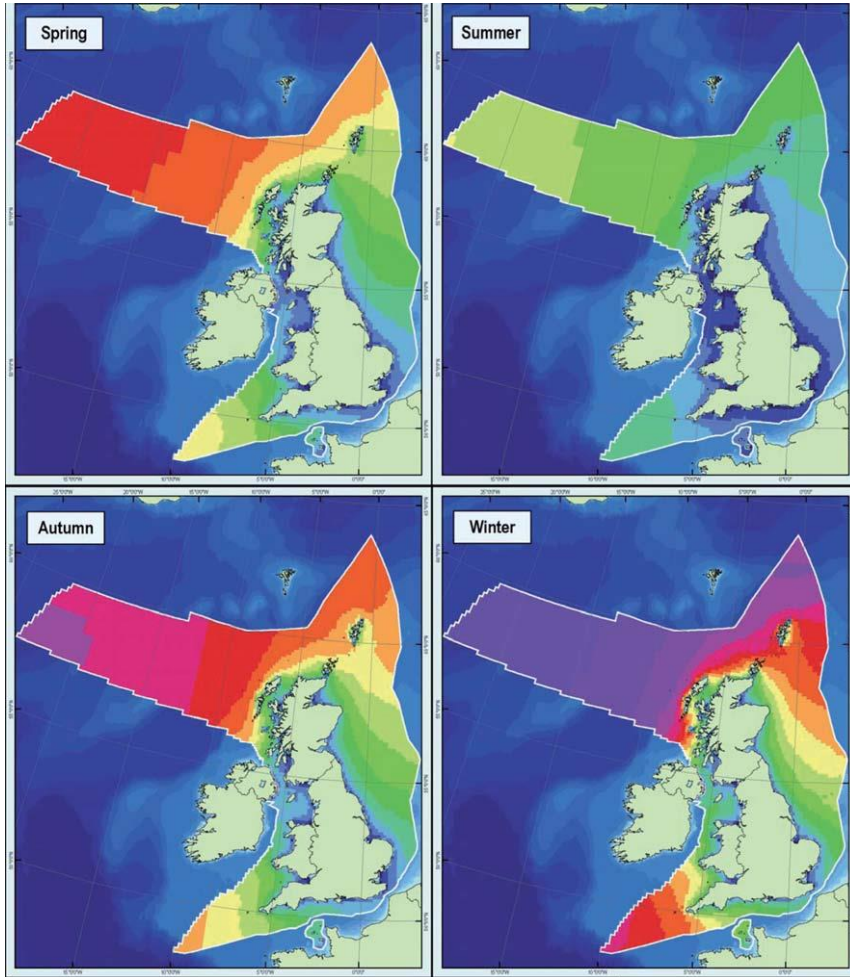
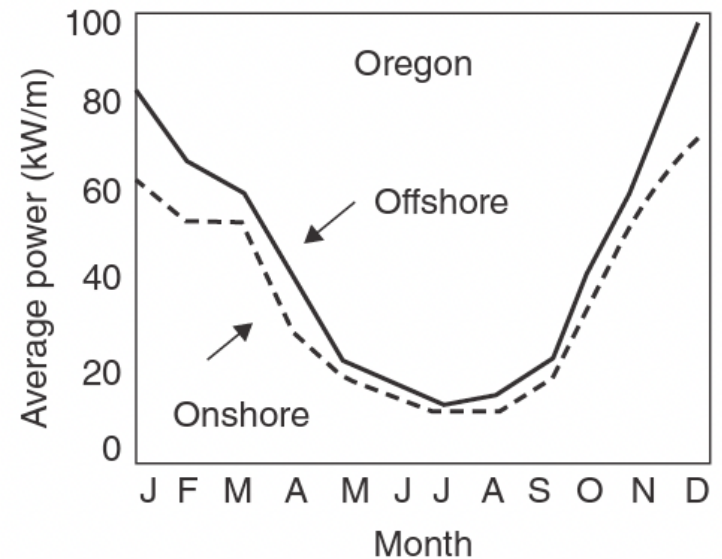
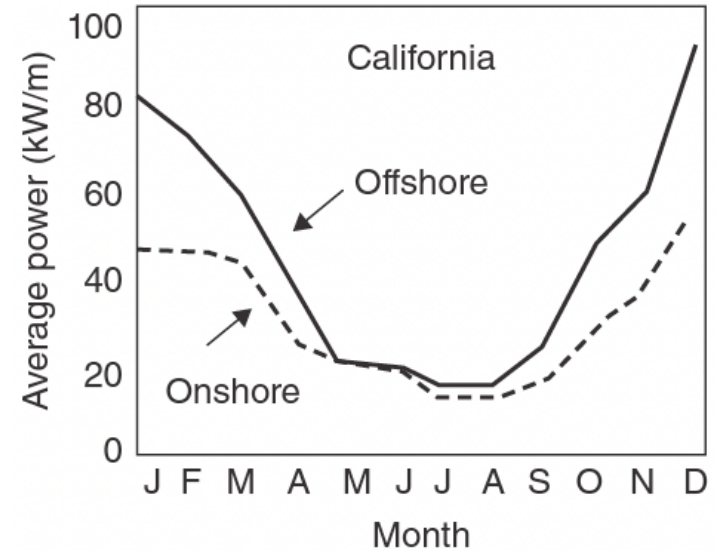


Fig. Area around the United Kingdom and the northeast Atlantic.



History of Wave Power

Year	Milestone	Key Contributor
1799	First known wave-energy patent filed in Paris	Pierre-Simon Girard & Son
1910	Early wave-powered system used to light a house in France	Bochaux-Praceique
1940s	Developed wave-powered navigation buoys using Oscillating Water Column (OWC) technology	Yoshio Masuda (Japan)
1974	Proposed the famous "Salter's Duck" wave-energy converter following the oil crisis	Stephen Salter (Univ. of Edinburgh)
1985	Construction of the TAPCHAN overtopping wave-energy plant in Norway	Norwegian Wave Power Program
2004	First commercial-scale offshore wave-energy farm using Pelamis technology*	Pelamis Wave Power (UK)

*Pelamis is a floating attenuator composed of hinged cylindrical sections that flex as waves pass.

Masuda's Navigation Buoy

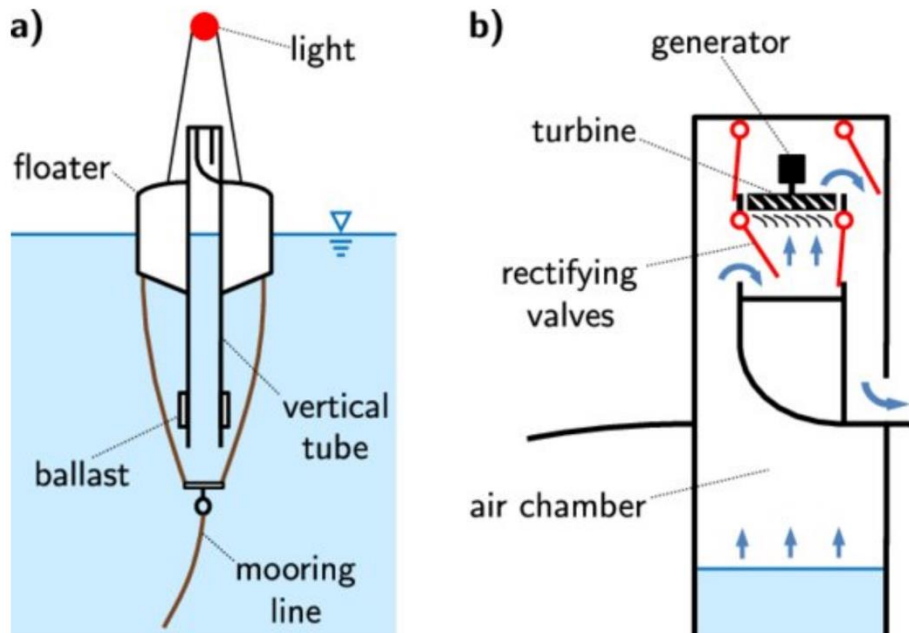
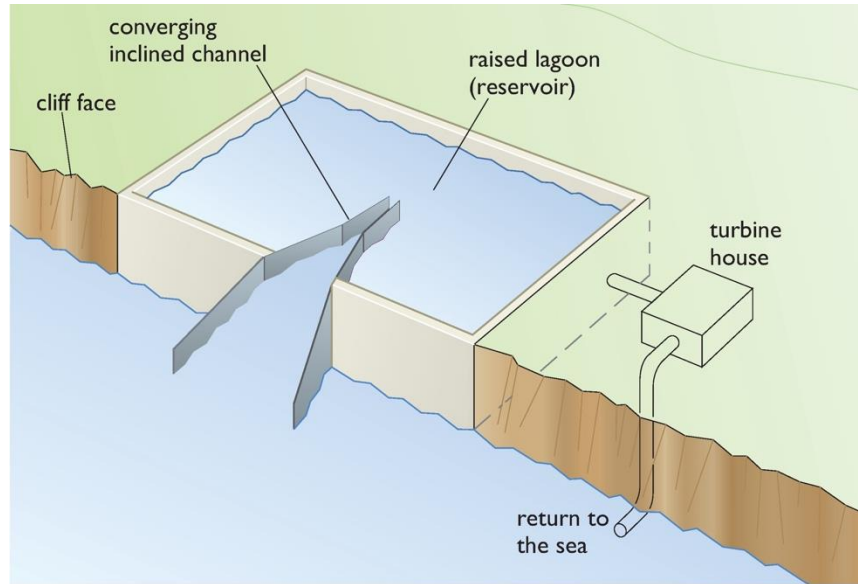


Fig: a) Yoshio Masuda's navigation buoy based on the OWC (oscillating water columns) principle.
b) Detail of the turbine and the rectifying valves of the PTO system under exhalation conditions.

- The device is a floater rigidly pierced by a **vertical tube**.
- The above the water line upper part forms an **air chamber** open to the atmosphere through a duct where an air turbine is installed.
- Wave action alternately **compresses/decompresses** the trapped air which forces air to flow through the turbine

Introductory Case



(a)



(b)

- A 350 kW prototype TAPCHAN wave energy converter built by Norwave in 1985 on a small Norwegian island.
- The channel walls 10m high and 170m long
- Kinetic energy in the waves → potential energy → electricity
- To be effective requiring a good wave climate:
 - high & consistent waves +
 - deep water close to the shore +
 - small tidal range

HOW OCEAN WAVES ARE FORMED

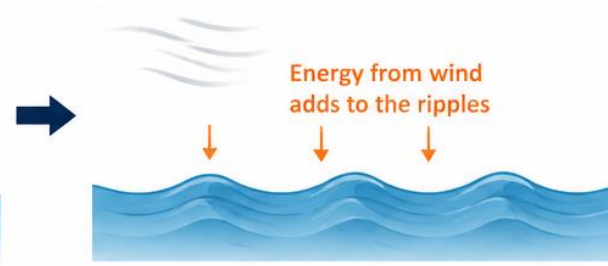
Wind transfers energy to the ocean surface, creating waves.

1 WIND BLOWS OVER WATER



Moving air exerts a shear (tangential) force on the water surface.

2 SMALL DISTURBANCES GROW



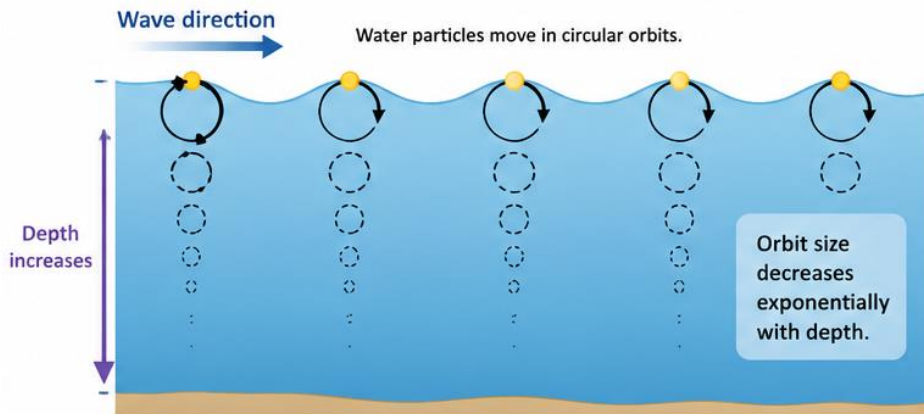
Turbulent airflow and pressure fluctuations add energy, making ripples grow into larger waves.

3 WAVES GROW AND TRAVEL



Wind pushes harder on the upwind face of waves, causing them to grow. Waves travel in the direction of the wind.

WHAT HAPPENS BENEATH THE SURFACE?



KEY TAKEAWAYS

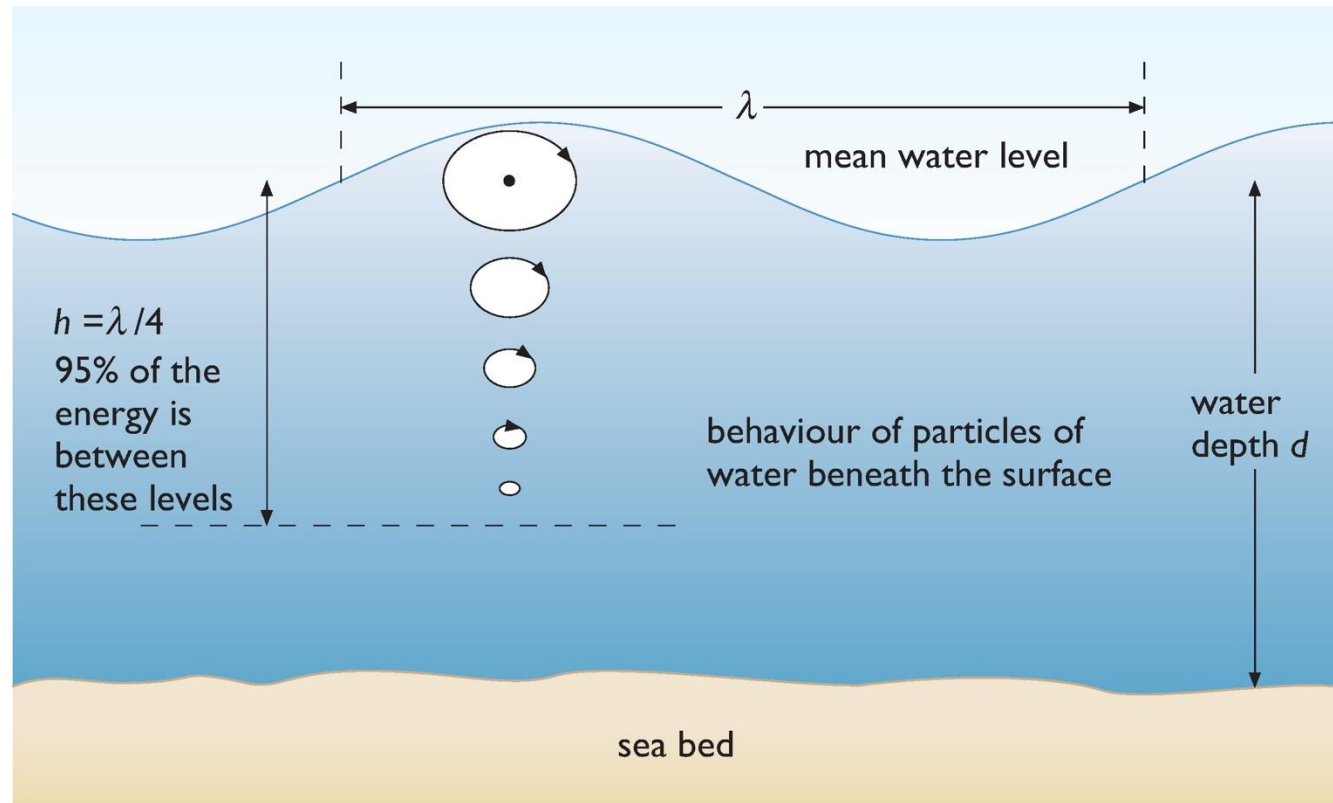
- Most wave energy comes from the Sun (drives wind).
- Wind energy is transferred to the ocean surface.
- Waves carry this energy across the ocean.
- Waves move energy, not water, forward.

FACTORS THAT AFFECT WAVE SIZE

- Wind speed**
(stronger = bigger waves)
- Fetch**
(longer distance = bigger waves)
- Duration**
(longer time = bigger waves)
- Depth**
(shallow water changes waves)

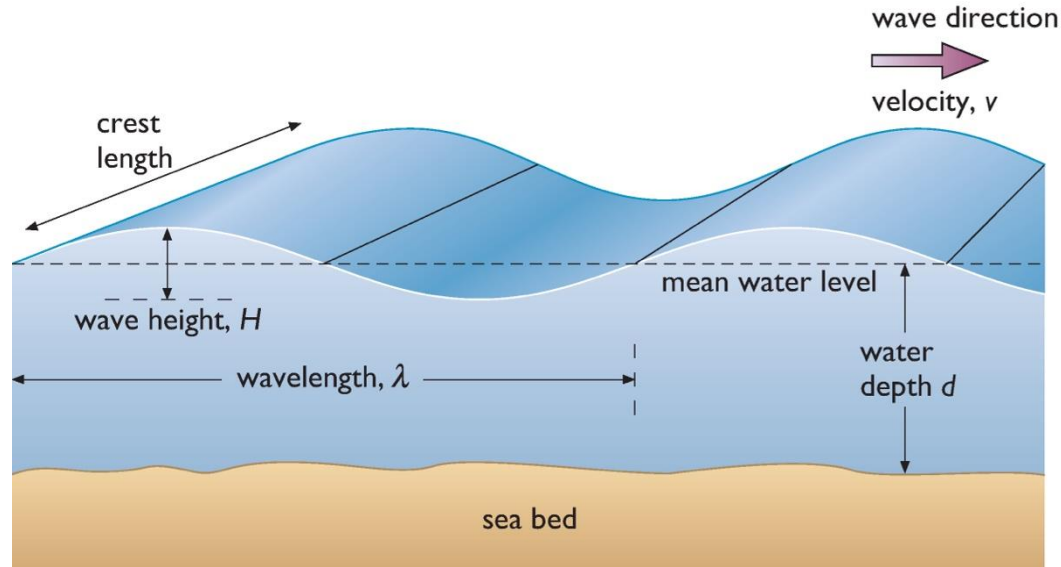


Beneath Surface

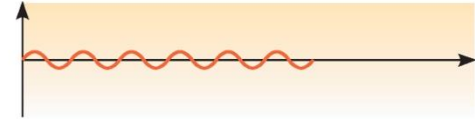


- Waves are composed of orbiting particles of water.
- Near the surface, orbits are the same size as the wave height.
- The size of orbits decreases exponentially with depth.

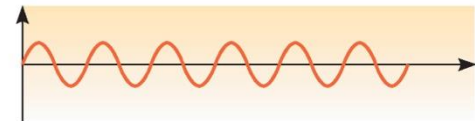
Key Characteristics of Wave



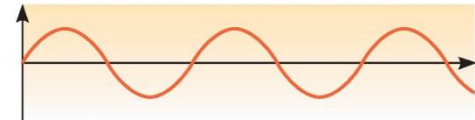
waves produced by a moderate wind



waves produced by a strong wind



waves produced by a strong wind over a long time period



The 'regular' wave is characterized by:

- **wavelength (λ)** – the distance between successive peaks
- **depth (d)** – the distance from mean water level to sea bed
- **height (H)** – the difference in height between peaks and valleys
- **period (T)** – the time in seconds taken for successive peaks to pass a given fixed point.

$$v = \lambda/T = \lambda f$$

Long waves travel faster than the shorter waves!

Wave Velocity



- Deep water waves ($d > \lambda/2$) : $v = \frac{gT}{2\pi}$
- Shallow water waves ($d < \lambda/4$) : $v = \sqrt{gd}$
- Intermediate water waves: $v = f(d, T)$

Frequency dispersion of gravity waves on the surface of deep water, shallow water and at intermediate depth, according to linear wave theory					
quantity	symbol	units	deep water ($h > \frac{1}{2} \lambda$)	shallow water ($h < 0.05 \lambda$)	intermediate depth (all λ and h)
dispersion relation	$\Omega(k)$	rad / s	$\sqrt{gk} = \sqrt{\frac{2\pi g}{\lambda}}$	$k\sqrt{gh} = \frac{2\pi}{\lambda} \sqrt{gh}$	$\sqrt{gk \tanh(kh)}$ $= \sqrt{\frac{2\pi g}{\lambda} \tanh\left(\frac{2\pi h}{\lambda}\right)}$
phase velocity	$c_p = \frac{\lambda}{T} = \frac{\omega}{k}$	m / s	$\sqrt{\frac{g}{k}} = \frac{g}{\omega} = \frac{g}{2\pi} T$	$\sqrt{gh}$	$\sqrt{\frac{g}{k} \tanh(kh)}$
group velocity	$c_g = \frac{\partial \Omega}{\partial k}$	m / s	$\frac{1}{2} \sqrt{\frac{g}{k}} = \frac{1}{2} \frac{g}{\omega} = \frac{g}{4\pi} T$	$\sqrt{gh}$	$\frac{1}{2} c_p \left(1 + \frac{2kh}{\sinh(2kh)}\right)$
ratio	$\frac{c_g}{c_p}$	-	$\frac{1}{2}$	1	$\frac{1}{2} \left(1 + \frac{2kh}{\sinh(2kh)}\right)$
wavelength	λ	m	$\frac{g}{2\pi} T^2$	$T\sqrt{gh}$	for given period T , the solution of: $\left(\frac{2\pi}{T}\right)^2 = \frac{2\pi g}{\lambda} \tanh\left(\frac{2\pi h}{\lambda}\right)$

Wave Power of Idealized Sinusoidal Wave

Using a classical result in linear wave theory, wave power per meter of wave crest P (kW/m) can be derived as

$$P = \frac{\rho g^2 H^2 T}{32\pi} \approx H^2 T$$

For example, a series of sinusoidal, 3-m waves with a period of 8 s would have power equal to

$$\begin{aligned} P &= \frac{\rho g^2 H^2 T}{32\pi} = \frac{1}{32\pi} \cdot 1025 \frac{\text{kg}}{\text{m}^3} \cdot \left(\frac{9.8 \text{ m}}{\text{s}^2} \right)^2 \cdot (3 \text{ m})^2 \cdot 8 \text{ s} \\ &= 70,500 \frac{\text{J/s}}{\text{m}} = 70.5 \text{ kW/m} \end{aligned}$$

- Hydropower (freshwater): $\rho \approx 998$
- Wave/Tidal energy (seawater): $\rho \approx 1025$

Wave Power of Actual Ocean Wave

Real Ocean Waves Are Irregular: consist of many wave components with different heights, wavelengths, and periods. Instead of analyzing each wave individually, engineers characterize sea states using two key parameters:

$$P = 0.42H_s^2T_p$$

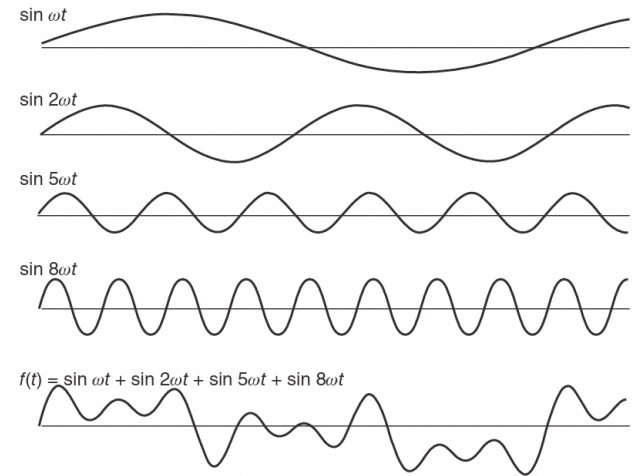
where:

H_s : **Significant Wave Height (m)**

- Average height of the *highest one-third of waves*
- Represents the overall "size" of the sea state
- Larger $H_s \Rightarrow$ significantly more wave energy

T_p : **Peak Wave Period (s)**

- Time between successive crests of the dominant waves
- Indicates how energetic the waves are
- Larger $T_p \Rightarrow$ more power carried by each wave



Typical energetic coastlines (California, Oregon, Scotland) have $H_s = 2 - 5m$ and $T_p = 8 - 15s$, yielding wave power densities of 20–100+ kW/m.

Wave Energy Converters (WEC)

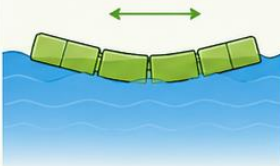
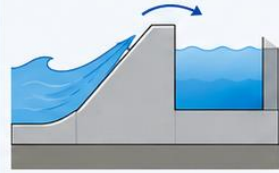
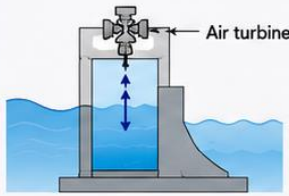
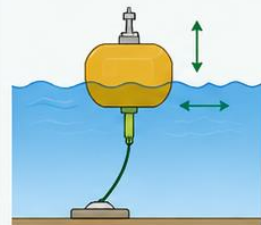


The goal of a wave energy converter: Convert the wave-induced motion into mechanical power that can drive an electrical generator.

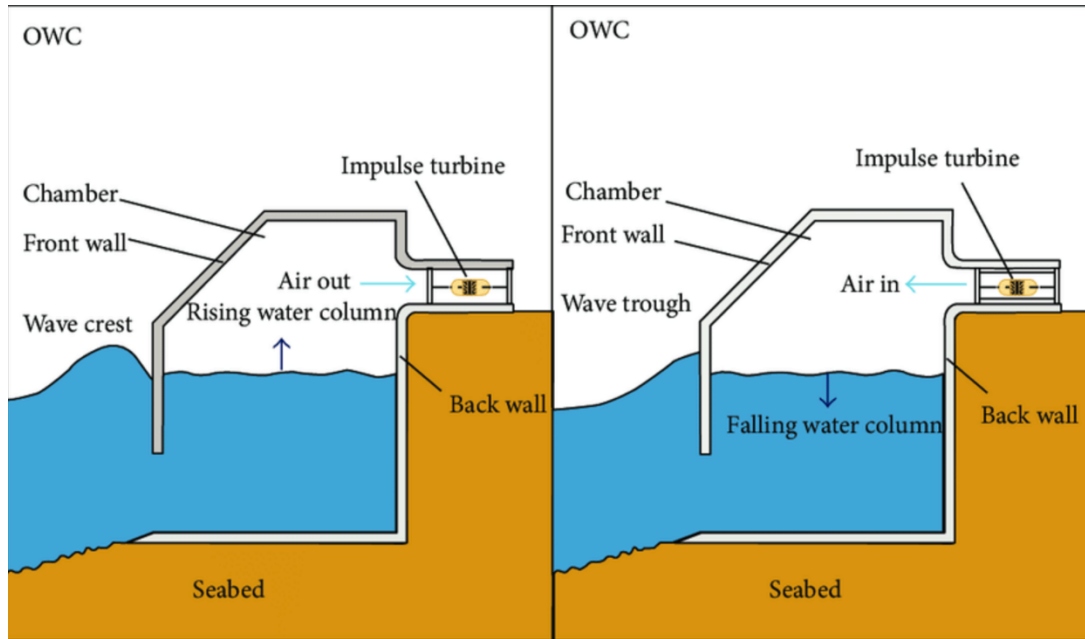
They are classified by: Mode of operation; device location; and geometry & orientation



TYPES OF WAVE ENERGY CONVERTERS

	1. Attenuators (Floating)	2. Overtopping (Terminator)	3. Oscillating Water Column (OWC)	4. Oscillating Wave Surge Converters (OWSC)	5. Point Absorbers	6. Others
Description						Various emerging devices (e.g., air compression, gyroscopic, flexible membranes) 
	Long floating structures aligned parallel to the wave direction. They absorb some of the wave energy through relative motion between segments.	Wave crest overtops a ramp and water is stored in an elevated reservoir. The stored water is released back to the sea through low-head turbines.	Water in a chamber rises and falls with waves, compressing and decompressing the air above it. The airflow drives a turbine.	Devices that extract energy from the horizontal surge motion of waves near the shoreline or seabed.	Floating bodies that move up and down (or in other directions) with waves. Motion is used to drive a power take-off system.	A variety of innovative converters that use air compression, gyroscopic motion, or flexible materials to extract wave energy.
Working Principle	- Wave passes along the structure - Differential motion between segments - Hydraulic/PTO system converts motion to power	- Waves overtop the ramp - Water accumulates in reservoir - Released through turbines - Generates electricity	- Water column oscillates - Air above is compressed and decompressed - Bidirectional airflow through turbine - Turbine drives generator	- Waves cause back-and-forth surge of device - Hydraulic or mechanical system resists motion - Energy captured and converted to electricity	- Device moves with waves - Motion drives a PTO (hydraulic, linear generator, etc.) - Electricity generated	- Use novel mechanisms (e.g., air compression, gyroscopic motion, flexible membranes) - Convert wave motion into electricity
Example (Real Device)	Pelamis 	TAPCHAN 	LIMPET 	WaveRoller 	PowerBuoy 	Ocean Grazer (Air Compression) 

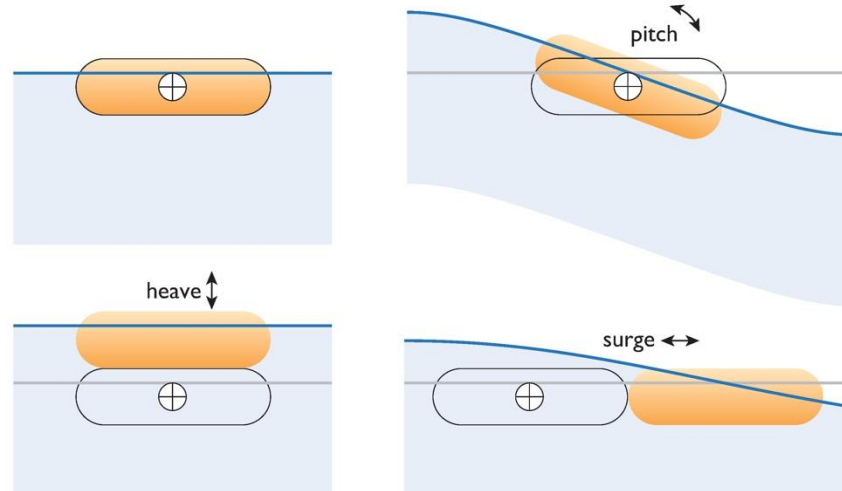
Oscillating Water Column (OWC)



- An OWC uses a large volume of moving water as a piston in a cylinder.
- Air is forced out of the column as a wave rises and is drawn in as the wave falls.
- This movement of air turns a weir turbine at the top of the column.

Harnessing Energy in Movements

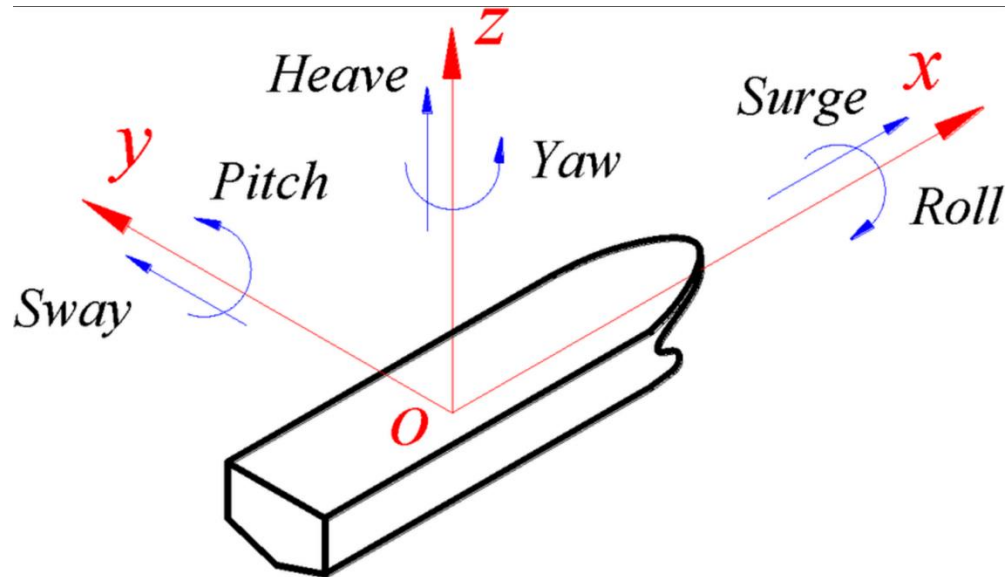
Wave-induced motion (**heave, pitch, or surge**) are leveraged



Motion	Description	Typical WEC	Energy Potential
Heave	Vertical up-and-down motion	Point Absorber	High
Pitch	Rotation about transverse axis	Attenuator (Pelamis)	High
Surge	Horizontal back-and-forth motion	WaveRoller, OWSC	Very High

Not Useful Movements

In wave energy conversion: *Sway, roll, and yaw* are not generally harnessed



Motion

Why Not Commonly Used?

Sway

Small lateral motion, difficult to extract energy

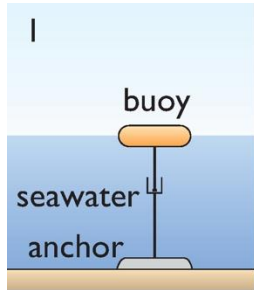
Roll

Irregular and highly device-dependent

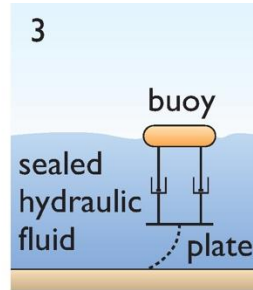
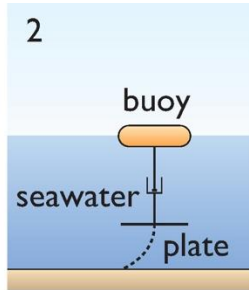
Yaw

Produces little useful relative motion

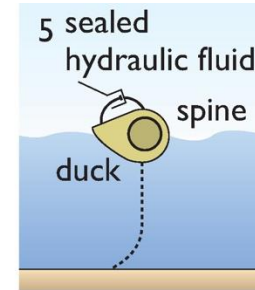
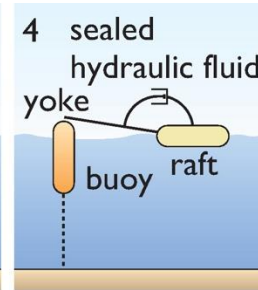
Mode of Operation



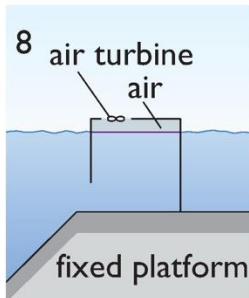
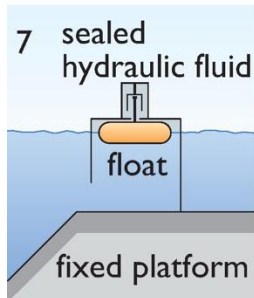
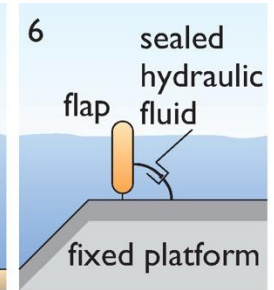
heaving float



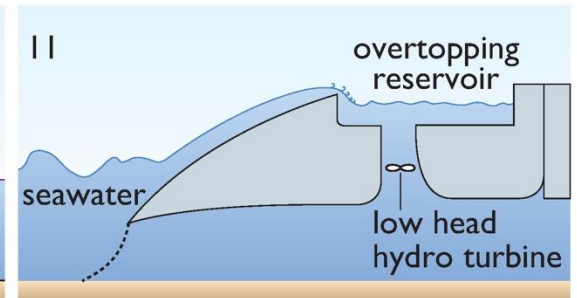
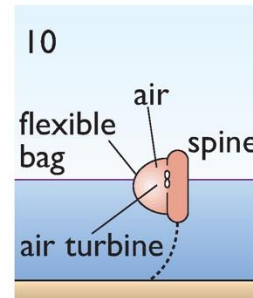
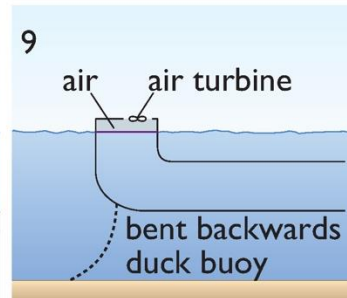
heaving and pitching float



pitching device

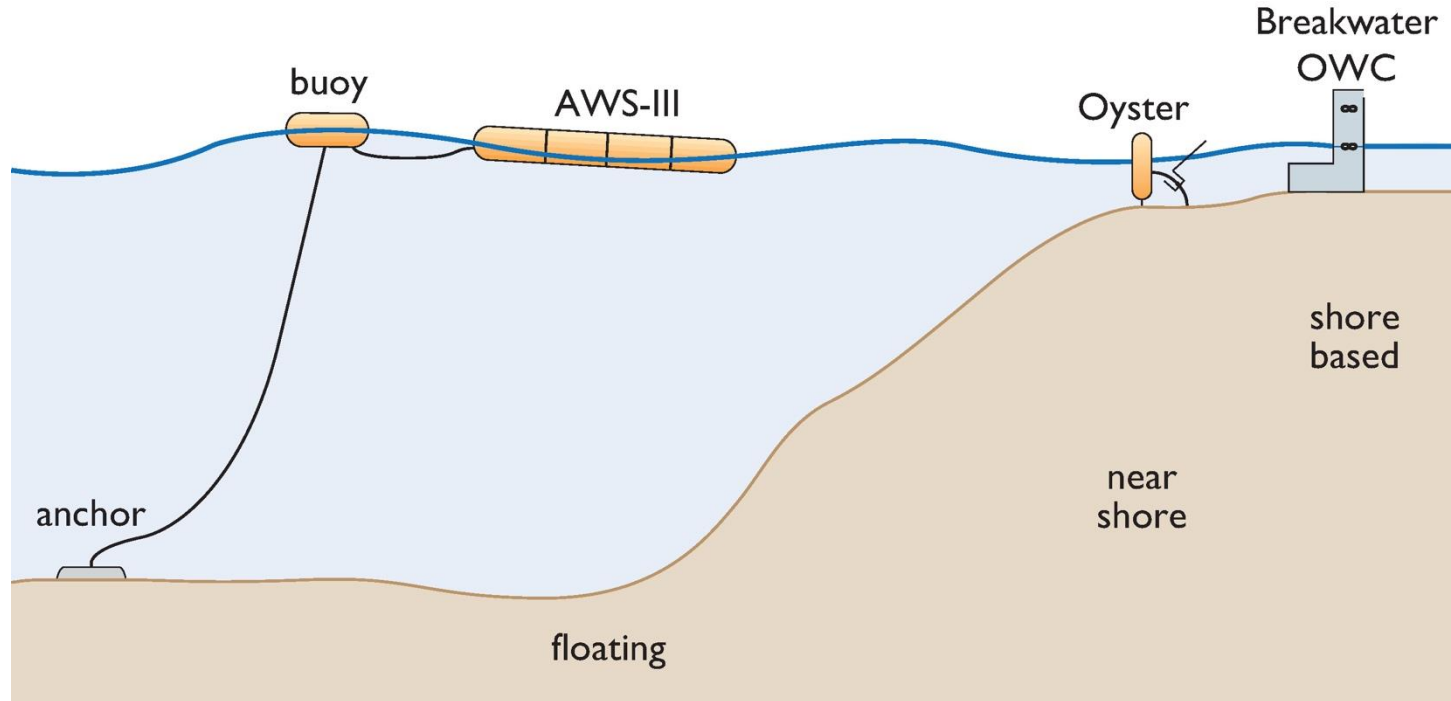


oscillating water column



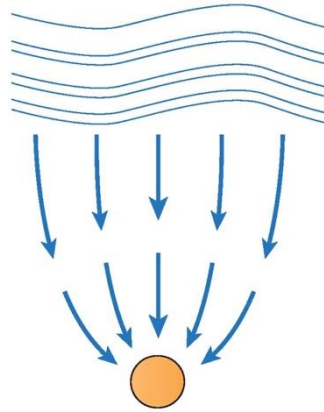
surge device

Device Location

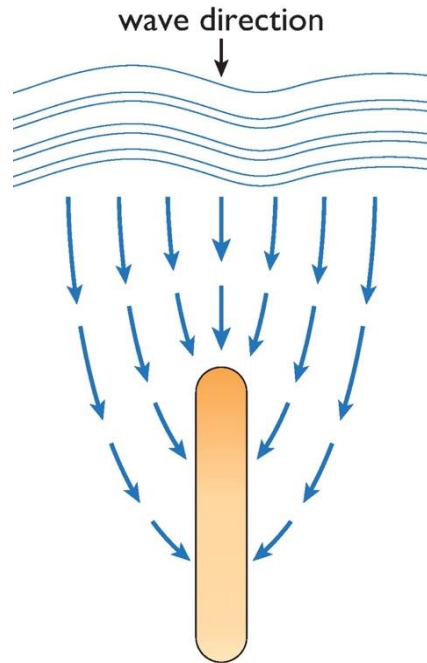


- fixed to the seabed, generally in shallow water (e.g. TAPCHAN)
- tie up in intermediate depths (e.g. Oyster)
- floating offshore in deep water (e.g. AWS-III).

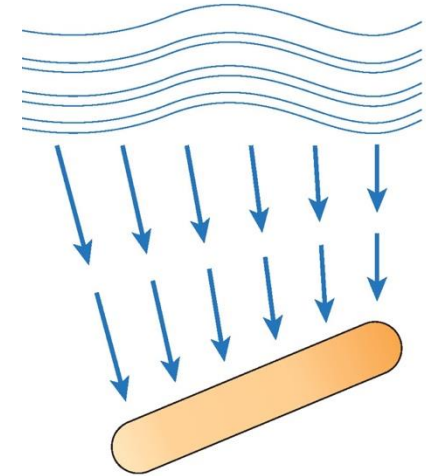
Geometry & Orientation



point
absorber



attenuator

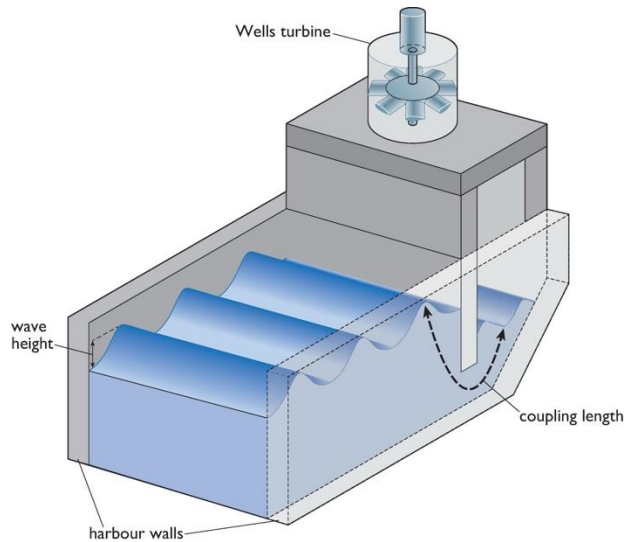


terminator
(at a slight angle to
improve performance)

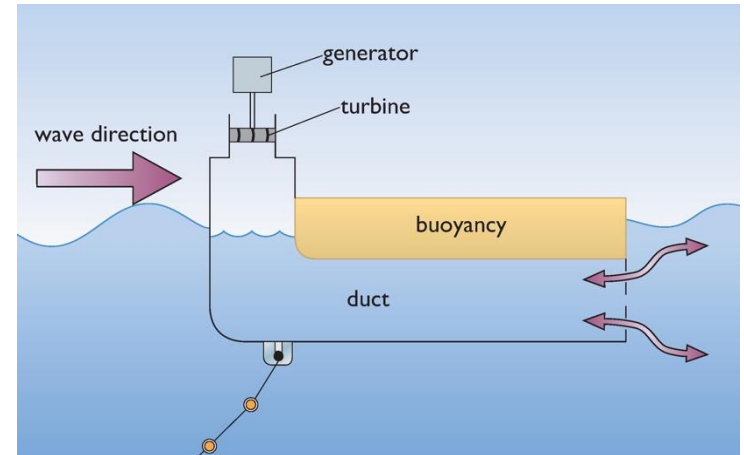
- Small dimensions, work by drawing wave energy from the water beyond their physical dimensions
- Principal axis perpendicular to the wave front
- Principal axis almost parallel to the incident wave front

Fixed vs Over-floating Devices (1/2)

Resonant oscillating water column (OWC)



Backward bent duct buoy (BBDB)



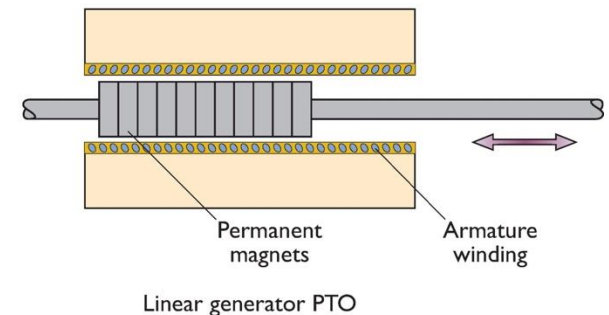
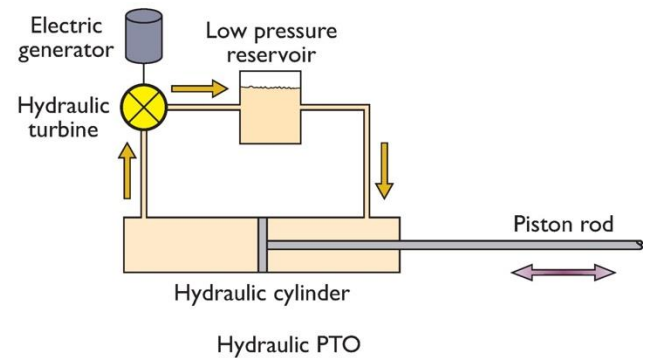
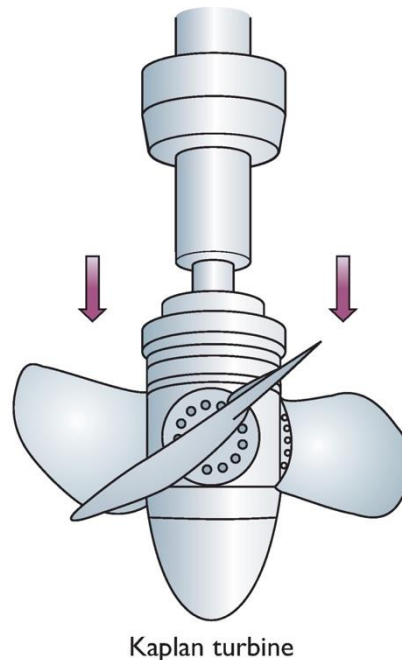
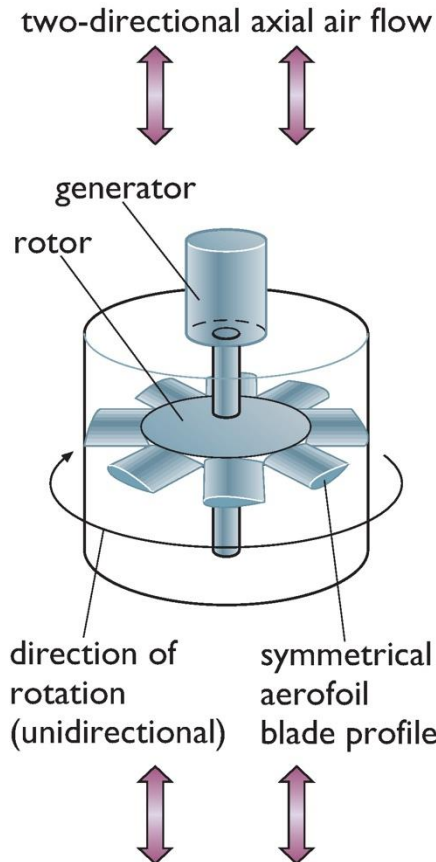
Fixed vs Over-floating Devices (cont'd)

Q: What advantages and disadvantages of fixed/shore mounted devices over floating ones?

Pros	Cons
Have a fixed frame of reference; closer to a grid	Generally operate in shallow water and hence at lower wave power levels
Good access for maintenance purposes	Geographical location – only a limited number of sites are suitable for deployment
The seabed attenuates storm waves that could otherwise destroy the device and turbine	To optimize output, they need to be positioned in an area of small <i>tidal</i> range

Power Take-off (PTO) Systems (1/2)

- The PTO is the "heart" of a wave energy converter, analogous to the turbine-generator set in a hydropower plant.
- Covert primary motions of waves into linear/rotary motions for an electricity generator

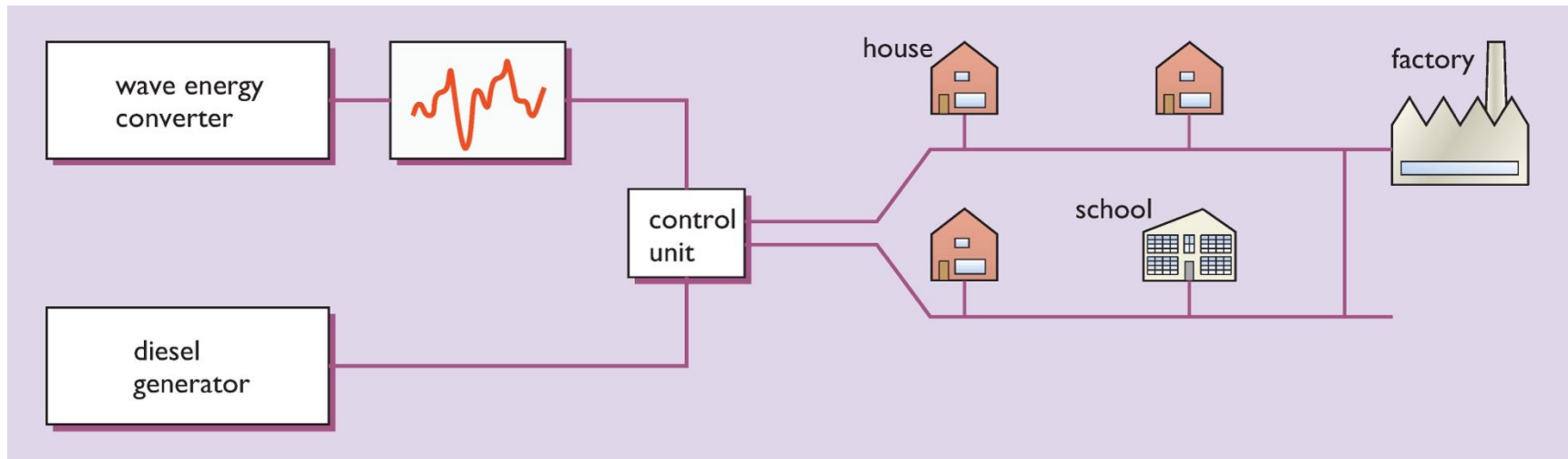


PTO Systems (2/2)

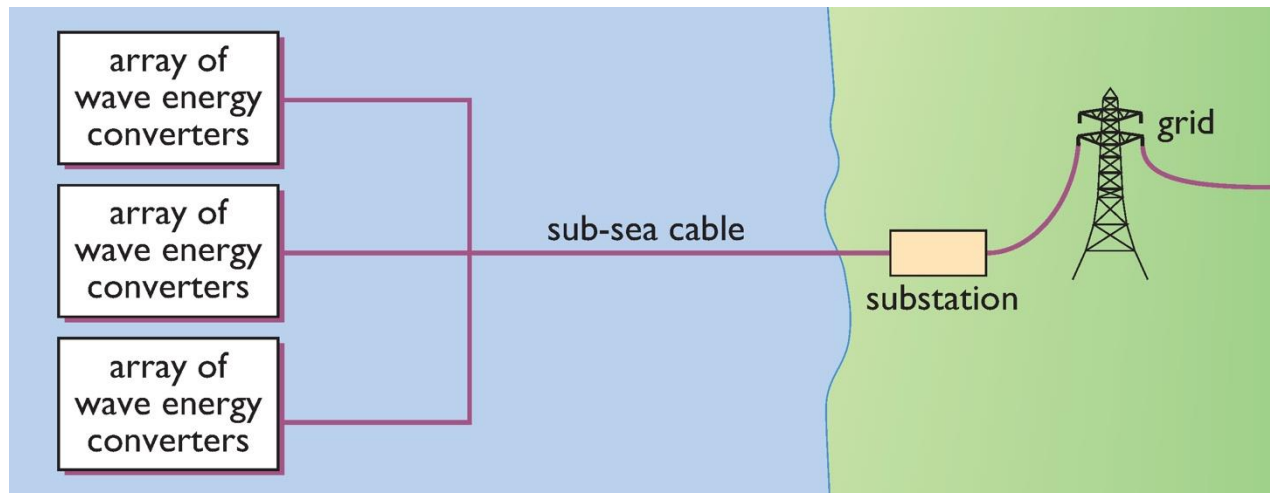
PTO Type	Principle	Example
Hydraulic	Fluid pressure	Pelamis
Air Turbine	Oscillating airflow	OWC
Linear Generator	Direct electromechanical conversion	Uppsala WEC
Mechanical	Gears and shafts	Early prototypes

Grid Integration

- For local communities



- For large power grids



Offshore Wind & Wave Energy

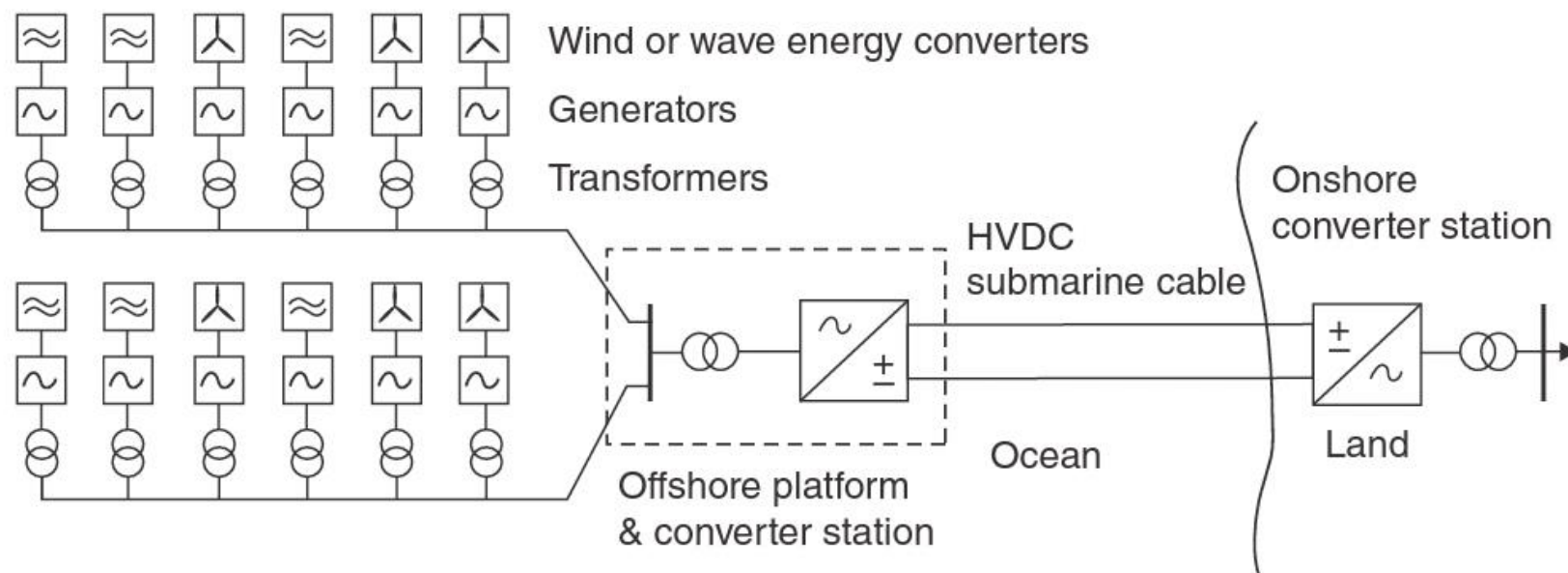
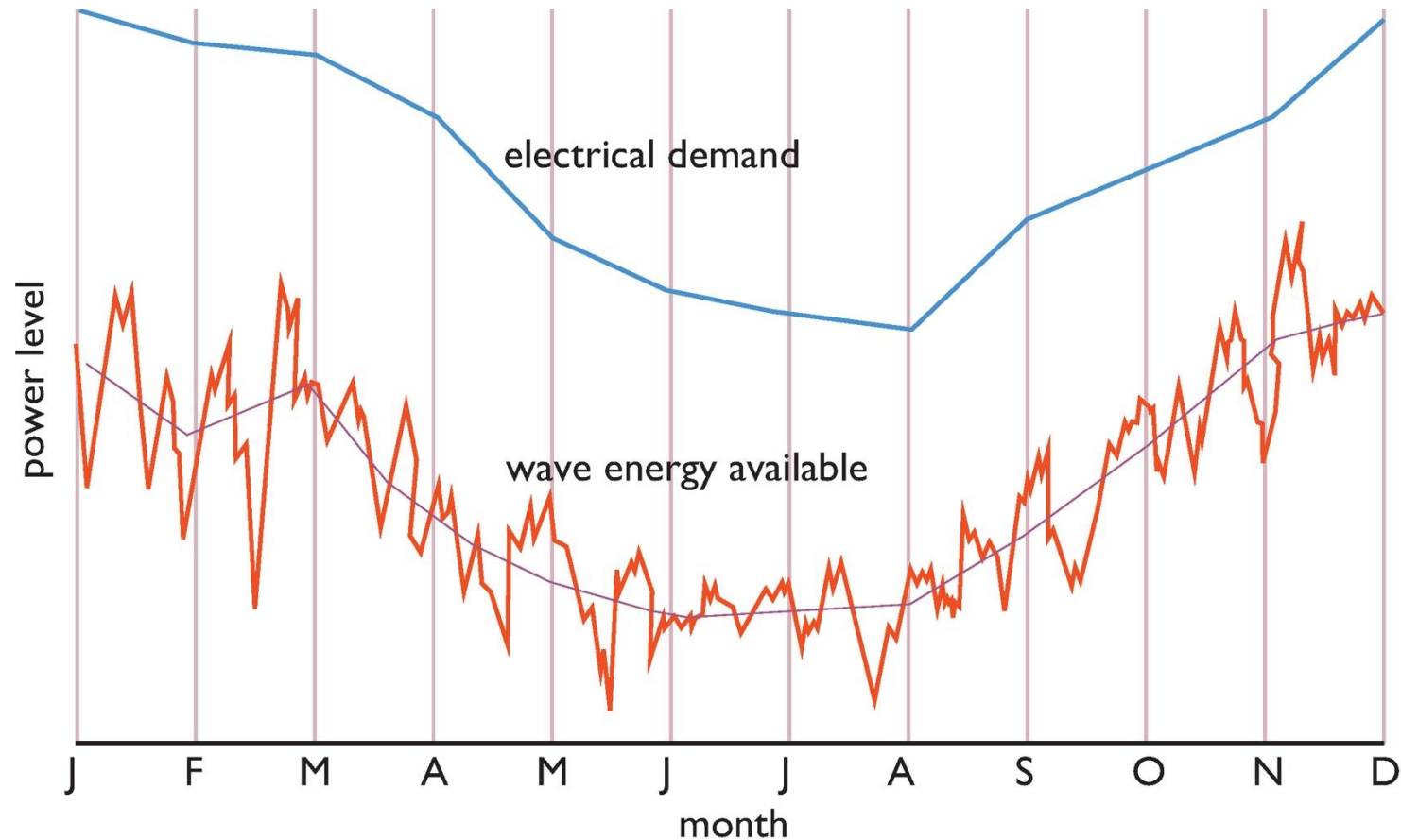


FIGURE 8.21 Colocated offshore wind and wave converters could share transmission costs. Redrawn from Stoutenburg and Jacobson (2011).

Seasonal Availability

Fig: Seasonal availability of wave energy and demand in UK



More wave energy units → smoother power level

Wave Energy Economics

	Wave energy station	Conventional station
Capital Cost	High	Low
Capacity Factor	Low	High

- Low CF (15%) the cost of electricity would be 29.1p kW/h
- High CF (22%) the cost of electricity would be 19.9p kW/h

The capital cost is around £3,000–4,500 per installed kW.

The cost of particular schemes may vary markedly from this.

Wave energy costs can only be competitive if the running costs are significantly below those for a conventional station.

Environmental Impact

- Little potential for chemical pollution. At most, they may contain some lubricating or hydraulic oil, which will be carefully sealed
- Little visual impact, except where shore-mounted.
- Noise generation is likely to be low – generally lower than the noise of crashing waves
- Present a small (though not insignificant) hazard to shipping.
- They should present no difficulties to migrating fish.
- Floating schemes are incapable of extracting more than a small fraction of the energy of storms so will not significantly influence the coastal environment.
- It is estimated that near-shore wave energy schemes will release very small amount of CO₂, SO₂ and NO_x

Summary

- Described the physical principles of wave energy
- Looked at the various wave energy resources available
- Introduced different wave energy technologies
- Studied wave energy economics
- Discussed the environmental impact of wave energy
- Examined the potential for grid integration of wave energy

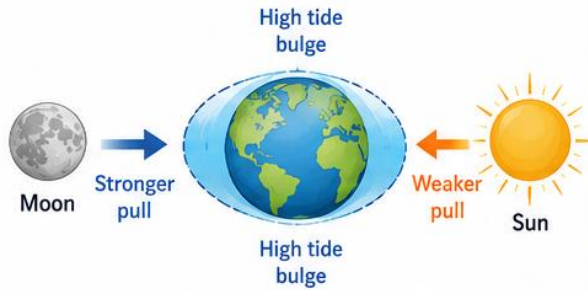
Ocean Energy - Wave Power Station

Tidal Energy

HOW TIDAL ENERGY WORKS

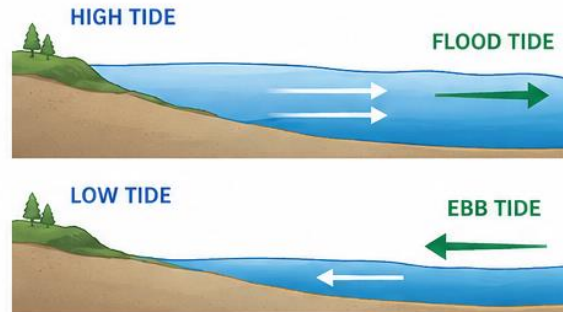
The gravity of the Moon and Sun creates predictable tides that can be converted into electricity.

1 GRAVITY CREATES TIDES



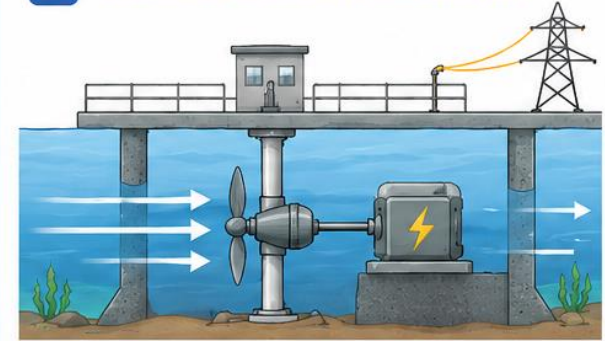
The Moon, and to a lesser extent the Sun, pull on Earth's oceans.

2 WATER LEVELS AND CURRENTS CHANGE



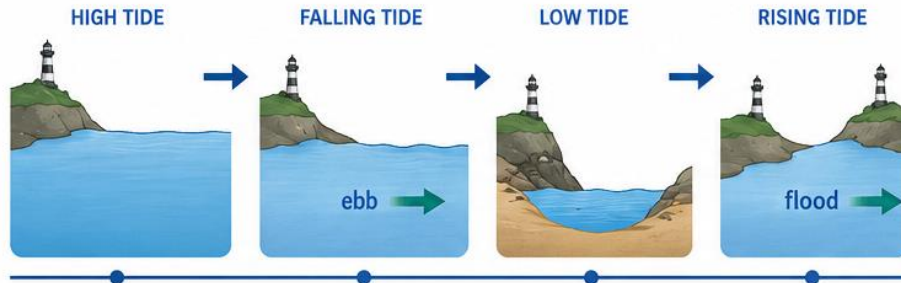
As sea level rises and falls, strong tidal currents are created.

3 TURBINES GENERATE ELECTRICITY



Moving water spins a turbine, which drives a generator to produce electricity.

WHAT HAPPENS DURING A TIDAL CYCLE?



Tides are highly predictable, typically with two high and two low tides each day in many coastal areas.

KEY TAKEAWAYS

- Tidal energy comes from gravity.
- Tidal currents are predictable.
- Turbines convert water motion into electricity.
- Best sites have strong currents and large tidal range.

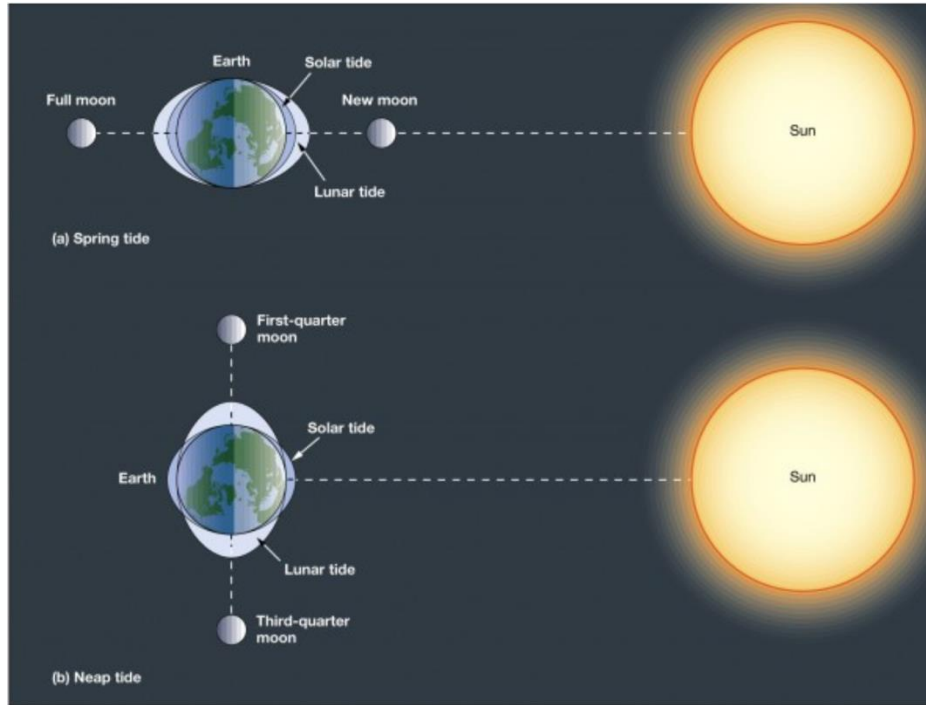
FACTORS THAT AFFECT TIDAL POWER

- Tidal range (larger range = more energy)
- Current speed (higher speed = more energy)
- Water depth (affects turbine design and output)
- Site/channel shape (constrictions increase currents)
- Turbine efficiency (better efficiency = more power)



Origin of the Tides

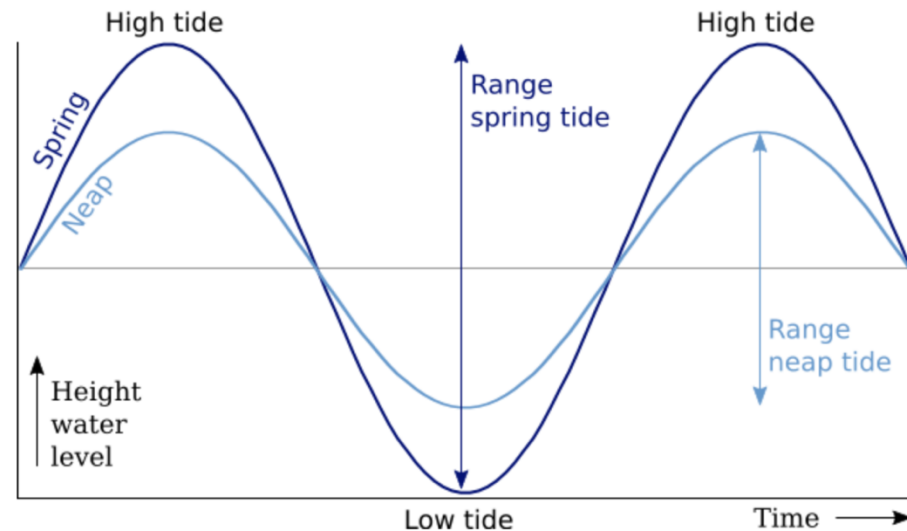
Tidal changes are due to gravitational forces exerted mostly by the moon, and partially by the sun.



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- Spring tides occur twice each lunar month all year long without regard to the season.
- Neap tides, which also occur twice a month, happen when the sun and moon are at right angles to each other.

Spring tide and neap tide levels are about 20% higher or lower than average.



Origin of the Tides (cont'd)

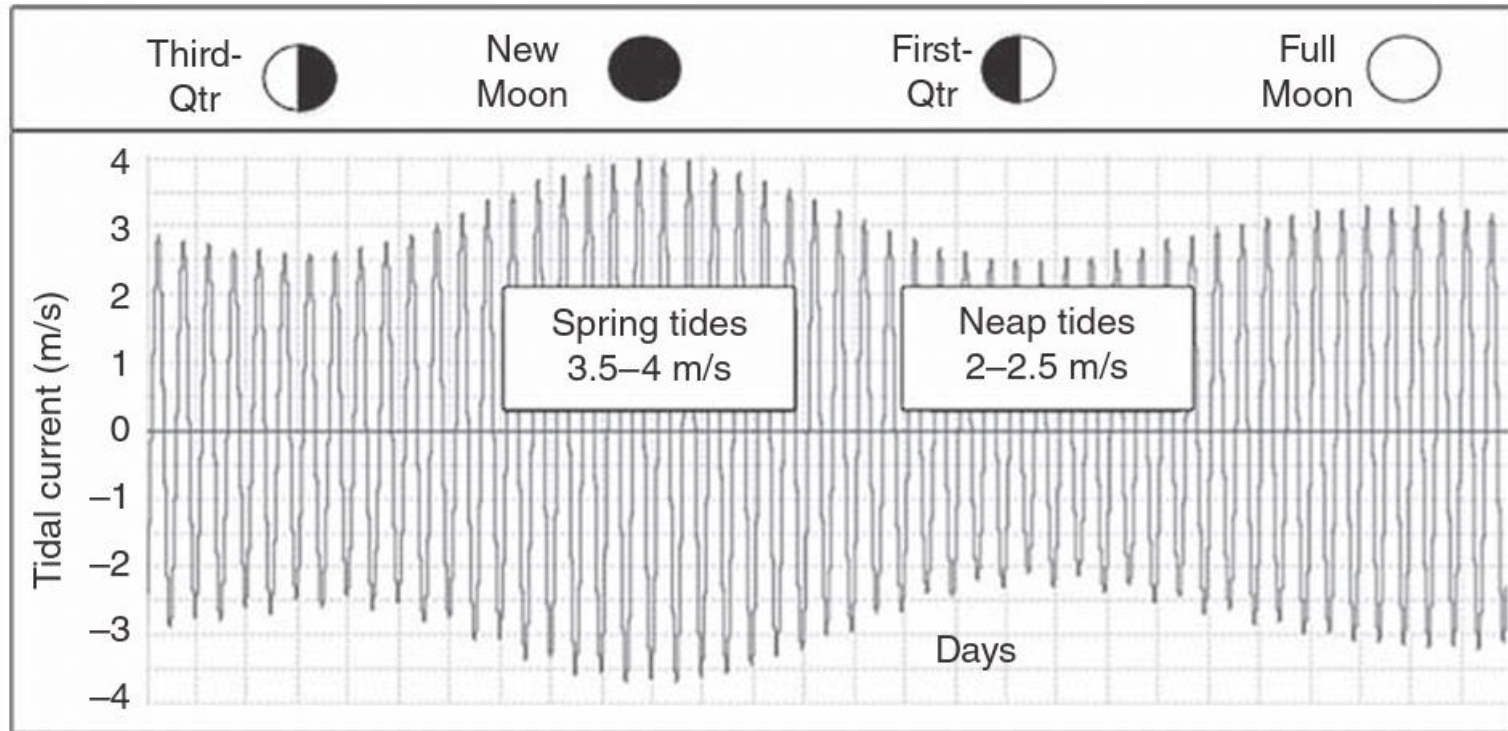


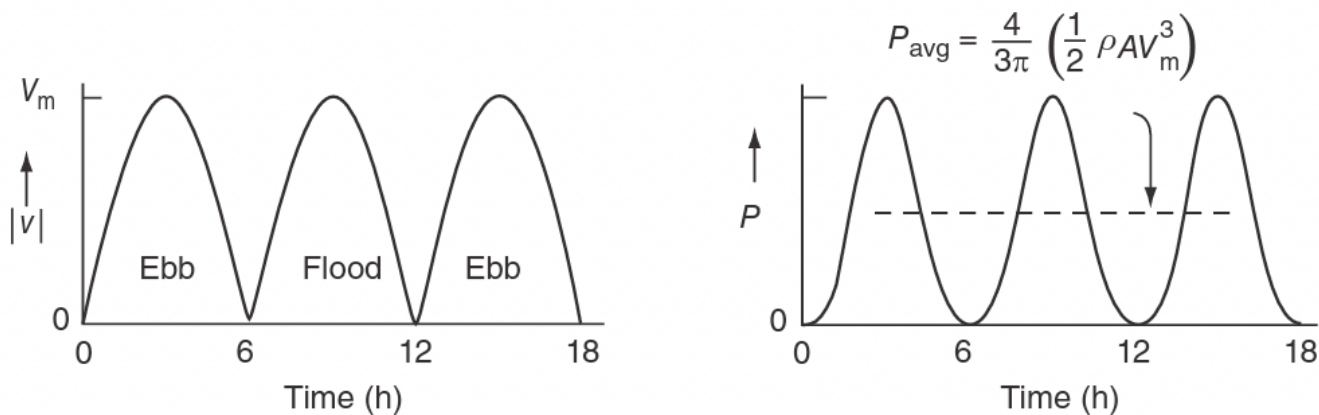
FIGURE 8.25 An example of the impact of spring and neap tides on tidal current.

Estimating In-stream Tidal Power

$$P_{\text{avg}} = \left(\frac{1}{2} \rho A v^3 \right)_{\text{avg}} = \frac{1}{2} \rho A (v^3)_{\text{avg}} \quad v = V_m \sin(\omega t)$$

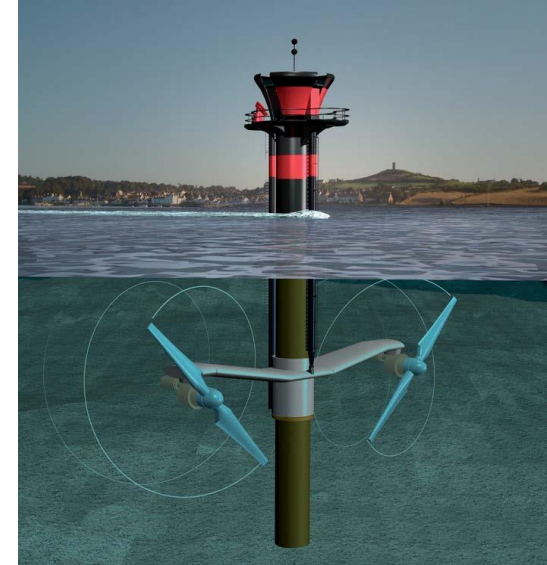
⇒ $(v^3)_{\text{avg}} = \frac{1}{T} \int_0^T f(t) dt = \frac{1}{\pi} \int_0^\pi (V_m \sin t)^3 dt = \frac{4}{3\pi} V_m^3$

⇒ $P_{\text{avg}} = \frac{1}{2} \rho A \frac{4}{3\pi} V_m^3 = \frac{2}{3\pi} \rho A V_m^3$



Tidal Energy Technologies

- Tidal energy include 1) kinetic energy obtained from the currents of changing tides; and 2) potential energy obtained from changing heights between the high and low tides.
- Technologies include tidal dams or barrages which contain a **sluice** across the water body. Beyond the sluice are hydro turbines. As the tide changes, the uneven water levels push through past the sluice and powers the turbine.



Advantages of Tidal Energy

- High efficiency of around 80%
- No GHG emissions
- Tidal stream generators have minimal environmental impact
- Low operations cost
- Reliable energy production (tides are constant & predictable)
- Very long lifespan (100+ years)

Read chapter 7 of textbook B3 !

[Tidal Power 101](#)

